Ruichen XU (徐睿晨)

Postdoctoral Researcher, University of Vienna

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Citizenship: People's Republic of China
Current Place of Residence: Vienna, Austria

Employment

- **University of Vienna** *Postdoctoral Researcher* Austria, Sept 2026 - Present
  - Mentor: Harald GROBNER, cosupervised by Adel BETINA,
  - Under the FWF Stand-Alone Grant project *Eigenvarieties and p-adic L-functions* (P 325085-N35).

Education

- **Chinese Academy of Sciences** *Ph.D student in mathematics* China, Sept 2021 - Jun 2026
  - Doctoral Advisor: Xin WAN.
  - Doctoral Thesis: *Iwasawa Theory for Unitary Groups via Eisenstein Congruences and its Applications.*
- **Beihang University** *Bachelor of Science in mathematics* China, Sept 2017 - July 2021
  - Undergraduate Advisor: Peng GAO and Shijin ZHANG (thesis advisor).
  - Undergraduate Thesis: *The Proof of Fermat's Last Theorem: Wiles' Proof and Beyond*

Publications and Preprints

◦ **Pseudonullity for fine Selmer groups over multiple $\mathbb{Z}_p^d$-extensions**
(With Peikai QI) Preprint (2026).
arXiv: 2608.25309.

✓ **On the Mordell-Weil rank of certain CM abelian varieties over anticyclotomic towers**.
(with Haidong LI) *Journal of Number Theory* 293 (2027) pp. 1-41.
DOI: 10.1016/j.jnt.2026.06.001 (open access). arXiv: 2502.12648.

◦ **The Gan-Gross-Prasad period of Klingen Eisenstein families over unitary groups**
Preprint (2024).
arXiv: 2410.13132.

Visiting Experiences

- **Université Marie et Louis Pasteur** *Visiting Doctoral Student* France, Dec 2024 - July 2025
  - Host Advisor: Olivier FOUQUET.
  - Funded by the program "*2024 International Research Collaboration Program for Postgraduate Students*" of Chinese Academy of Sciences (1 350 EUR monthly).
- **Peking University** *13th Enhanced Program for Graduate Study* China, Spring 2021
- **Peking University** *12th Enhanced Program for Graduate Study (Online)* China, Spring 2020

Conference Talks

- **33èmes Journées Arithmétiques** July 1, 2025
  Title: The Gan-Gross-Prasad period of Klingen Eisenstein families over unitary groups.
  Location: Université du Luxembourg. Esch-sur-Alzette, Luxembourg.

- **Géométrie algébrique et théorie des nombres en Bourgogne Franche-Comté** Mar 28, 2025
  Title: The method of Eisenstein congruences in Iwasawa theory.
  Location: Institut de Mathématiques de Bourgogne. Dijon, France.

Teaching Experiences

- **Advanced Abstract Algebra** *Teaching Assistant, at Beihang University* Fall 2020
  Undergraduate course on Galois theory, the theory of modules and the representation theory.
- **Complex Analysis** *Teaching Assistant (Online), at Beihang University* Spring 2020
  Undergraduate course on basic complex analysis for students majoring in mathematics.
- **Undergraduate Mentors Program** *Mentor, at Beihang University* Fall 2018 - Spring 2019
  Guiding freshmen in making a smooth transition from middle school to undergraduate studies by providing psychological and administrative support, along with academic guidance in foundational courses including calculus, mathematical analysis, linear algebra, programming (C/Python), and physics.

Scholarships

- **Excellent Award of the Dean's Scholarship** *AMSS, CAS* 2025
  for outstanding graduating students at the Academy of Mathematics and Systems Science (AMSS), Chinese Academy of Sciences (CAS). (22 000 CNY).
- **Professor Hua Loo-Keng Scholarship** *AMSS, CAS* 2023, 2024
  for excellence in PhD qualifying exams and research (100 000 CNY each).
- **Professor Hua Loo-Keng Scholarship** *AMSS, CAS* 2019, 2020, 2021
  for excellence in undergraduate mathematical coursework and contests (2 000 CNY each).

Conferences Attended

- **2026 Jinling Conference of Number Theory** *Nanjing, Jiangsu, China* Apr 2026
  At *Nanjing University.*
- **Relative Langlands and Arithmetic** *Marseille, France* Feb 2026
  At the *Centre International de Rencontres Mathématiques.*
- **Conference in honor of Professor George Lusztig** *Hong Kong SAR, China* Nov 2025
  At the *University of Hong Kong.*
- **33èmes Journées Arithmétiques** *Esch-sur-Alzette, Luxembourg* July 2025
  At the *Université du Luxembourg.*
- **Iwasawa Theory and Representations of p-Adic Groups** *Lille, France* June 2025
  At the *Laboratoire Paul Painlevé, Université de Lille.*
- **Géométrie algébrique et théorie des nombres en Bourgogne Franche-Comté** Mar 2025
  At the *Institut de Mathématiques de Bourgogne*, *Dijon, France.*
- **Young researchers in Galois Representations and related topics** *Genova, Italy* Jan 2025
  At the *Università di Genova.*
- **Workshop on p-adic Geometry** *Singapore* Nov 2024
  At the *National University of Singapore.*

- **BSD Conjecture and Goldfeld Conjecture** *Kunming, Yunnan, China* Oct 2024
  At *Tianyuan Mathematics Research Center.*
- **Development of Iwasawa Theory** *Yokohama, Kanagawa, Japan* Jul 2024
  A conference in honor of the 60th birthday of Masato Kurihara. At *Keio University.*
- **2024 Jinling Conference of Number Theory** *Nanjing, Jiangsu, China* Apr 2024
  At *Nanjing University.*
- **Special Values of L-functions** *Sanya, Hainan, China* Jan 2024
  At *Tsinghua Sanya International Mathematics Forum.*
- **The 3rd Cross Strait Workshop on Arithmetic Geometry** *Hefei, Anhui, China* Aug 2023
  At *the University of Science and Technology of China.*
- **Workshop on Arithmetic and Topology** *Shenzhen, Guangdong, China* Aug 2023
  At *the Southern University of Science and Technology.*
- **Iwasawa 2023: in memory of John Coates** *Cambridge, the United Kingdom* July 2023
  At *the University of Cambridge.*
- **Satellite Conference in Number Theory of ICBS** *Beijing, China* July 2023
  At *Morningside Center of Mathematics, Chinese Academy of Sciences.*
- **2023 Jinling Conference of Number Theory** *Nanjing, Jiangsu, China* June 2023
  At *Nanjing University.*

Summer Schools Attended

- **Summer school on Galois representations and Reciprocity** *Hanoi, Vietnam* July 2024
  At *Vietnam Institute for Advanced Study in Mathematics.*
- **2021 Summer School on Number Theory** *Xi'an, Shaanxi, China* July 2021
  At *Xi'an Jiaotong University.*
- **Theta Correspondence and Its Arithmetic Applications** *Suzhou, Jiangsu, China* July 2021
  At *Soochow University.*
- **Short Courses on Number Theory** *Xiamen, Fujian, China (Online)* July 2020
  At *Xiamen University.*
- **2020 SCMS Algebraic Geometry Summer School** *Shanghai, China (Online)* July 2020
  At *Shanghai Center of Mathematical Sciences, Fudan University.*
- **"The Road to Mathematics" Int'l Summer School** *Harbin, Heilongjiang, China* July 2019
  At *Harbin Institute of Technology.*

Seminars Organized and Attended

As the organizer and the main speaker of the following seminars among students:

- **Universal Iwasawa theory** Fall 2024
  Study the the article "*The Iwasawa Main Conjecture for universal families of modular motives*" by O.Fouquet and X.Wan.
- **Automorphic methods in Iwasawa theory** Spring 2023 - Fall 2023, Spring 2025
  Study the machinery of Eisenstein congruences via the article "*The Iwasawa Main Conjectures for* GL_2" by C.Skinner and E.Urban, and its later developments.

- **Iwasawa Theory of elliptic curves** Fall 2021, Fall 2023
  Study the article "*Introduction to Iwasawa Theory for Elliptic Curves*" by R.Greenberg, with a special focus on solving all the exercises.
- **Automorphic Forms and Automorphic Representations** Spring 2022 - Fall 2022
  Following the textbook "*An Introduction to Automorphic Representations*" by J.Getz and H.Hahn.

As the organizer and a speaker of the following seminars among students:

- **Rational points on elliptic curves** Fall 2024
  Following the textbook "*Rational Points on Modular Elliptic Curves*" by H.Darmon.
- **Arithmetic of elliptic curves** Fall 2023 - Spring 2024
  Following the textbooks "*Arithmetic of Elliptic Curves*" and "*Advanced Topics in the Arithmetic of Elliptic Curves*" by J.Silverman.

Languages

- **Chinese**: Native
- **English**: Proficient
- **French**: Beginner

(Updated: Sept 2, 2026)